OpenMMLA: an IoT-based Multimodal Data Collection Toolkit for Learning Analytics

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Abstract

Multimodal Learning Analytics (MMLA) expands traditional learning analytics into the digital and physical learning environment, using diverse sensors and systems to collect information about education in more real-world environments. Challenges remain in making these technologies practical for capturing data in authentic learning situations. With the advent of readily accessible powerful artificial intelligence that includes multimodal large language models, new opportunities are available. However, few approaches allow access to these technologies, and most systems are developed for specific environments. Recent work has begun to make toolkits with access to collecting data from sensors, processing, and analyzing, yet these tools are challenging to integrate into a system. This paper introduces OpenMMLA, a toolkit approach that provides programming interfaces for harnessing these technologies into an MMLA platform with prebuilt pipelines, including the audio analyzer, indoor positioning, and video frame analyzer, offering multimodal data collection and visualizations and analytics. The paper provides an initial evaluation of the functionalities of the toolkit in data capturing and the implemented pipelines' performances.

CCS Concepts

• **Human-centered computing → Collaborative and social computing**.

Keywords

Multimodal Learning Analytics, Group Work, Internet of Thing, Smart Badges

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1 Introduction

Over the last decade, research in Multimodal Learning Analytics (MMLA) has become a field within Learning Analytics (LA). MMLA combines many disciplines to investigate and support learning processes using diverse sensors, data streams, and systems for collection and analysis [22, 36]. However, practical challenges such as providing easier-to-use systems deployed in diverse and complex learning situations to capture learning in authentic situations [4], the infrastructure and costs of the equipment, the performance of the system in real-world settings that include diverse lighting, high levels of audio, along with reliability and generalisability, underscore the need for further research in how to develop systems and technologies for this area [5].

Now, the rapid development of powerful and accessible artificial intelligence (AI) frameworks (e.g., "OpenAI, Azure, Meta, Whisper, Llama") allows for the development of new tools that, in real-time, can transcribe speech-to-text and describe and analyze physical interaction between people through interpreting video [37]. These tools offer new methods for capturing and analyzing multimodal data across physical and digital learning environments with new forms of interaction and feedback for learners and teachers that utilize AI [7, 30]. Designing, developing, and deploying new MMLA systems that allow for emerging technologies to be used across different learning contexts has been an elusive aim for researchers. Currently, the need for available systems and the generalizability of how these platforms and software perform in diverse and authentic learning environments have impeded the adoption of these tools [6]. Developing a system incorporating hardware, software, and AI that considers the context and complexity of education and the vulnerable nature of children, learners, and teachers requires different approaches. A key challenge for MMLA is developing general platforms that support diverse collaboration across various contexts. These platforms must be open, adaptable, flexible, usable, and affordable for researchers and practitioners. This requires tools (sensors, data pipelines, inference) based on collaborative learning theories, leveraging interaction modalities like speech, movement, proximity, and behavior, all within a configurable platform. The research aim is: *How can we design and implement an MMLA toolkit that enables the development of sensor-based data collection pipelines and analytics systems to study collaboration in small group settings effectively?*

2 Related Work

2.1 Collaborative Problem Solving and Group Work

MMLA research has primarily explored collaborative problem-solving (CPS) through computer-supported collaborative learning

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(CSCL) principles [9, 28]. Understanding and supporting CPS's cognitive and social aspects remains a focus of education [13]. MMLA has studied small group work in various educational settings, capturing behavioral interactions and linking them to emotions, motivation, and cognition [12, 32]. Current systems have progressed in tracking low-level interactions such as engagement, gaze, turn-taking, and physiological signals, as well as digital learning interactions [1, 16]. Recent studies emphasize the importance of audio analysis for understanding speech patterns, turn-taking, and engagement [24, 25]. Advances in machine learning now enable real-time speech-to-text and discourse analysis, offering insights into learners' behaviors and interactions [3].

2.2 Multimodal Data Collection Tools and Platforms for LA

The landscape of MMLA platforms is split between research-based and commercial systems, each with distinct advantages and limitations. The Social Signal Interpretation Framework (SSI) and Platform Situated Intelligence (PSI) cater to various uses, including robotics and mixed reality [2, 33]. Other academic projects like the Multimodal Learning Hub [29], OPAF [21], and Kumitron [11] leverage multimodal data to explore learning environments. CoTrack analyzes group work through audio and video [6] and provides collaboration feedback. Sociometric wearable devices (SWDs) like Rhythm and Open Badges collect interaction data [18], while Yamaguchi's Sensor-based Profiler tracks collaboration phases via chest sensors [35]. SWDs offer data control benefits, avoiding heavy reliance on facial recognition. Commercial platforms like Noldus and iMotions focus on behavioral observation but are costly and not tailored to education [27, 38].

EZ-MMLA Toolkit takes a novel approach and creates a toolkit for researchers to investigate using the latest machine learning algorithms to collect diverse data streams from webcams: attention (eye-tracking), physiological states (heart rate), body posture (skeletal data), hand gestures, emotions (from facial expressions and speech), and lower-level computer vision algorithms (e.g., fiducial and color tracking) [14]. The EZ-MMLA Toolkit presents a novel approach toward providing more flexible tools for MMLA research. Still, it is designed primarily for teaching and demonstrations and does not provide a way to create an integrated platform. The challenge for MMLAs that support CPS and group work is identifying the correct interactions in group dynamics for data collection and developing tools and platforms that can work across contexts. We see a research opportunity to design and develop a dual approach that divides the system into two parts: a toolkit and a platform. This paper focuses on developing such a toolkit as a basis for creating diverse platforms to capture collaboration across contexts.

3 OpenMMLA

3.1 High-level Overview

The OpenMMLA harnesses multimodality to enhance learning analytics through increased data diversity, ensuring that each modality complements others, bringing added value that cannot be obtained from other modalities alone.

At its core, OpenMMLA[1] includes:

- **OpenMMLA Toolkit:** An open-source software package that enables developers to design and implement customized learning analytics pipelines. This toolkit comprises a suite of *streams*, *bases*, *services*, and *utils* functions, etc.
- **OpenMMLA Platform:** An MMLA platform integrated with built-in pipelines from the toolkit for researchers to customize and collect multimodal data at various levels (individual/group, raw/features). It also gives teachers and students insights through visualizations and analytical results displayed on a shared dashboard.

3.2 Design Considerations

The toolkit design focuses on providing modular and extensible interfaces for MMLA pipeline development. At the same time, the platform follows key design principles that address MMLA concerns [8] and module requirements [19], including data privacy, flexibility, and scalability. These principles are implemented through a multilayered Internet of Things (IoT) architecture and client-server architecture, as illustrated in Figures 1 and 2.

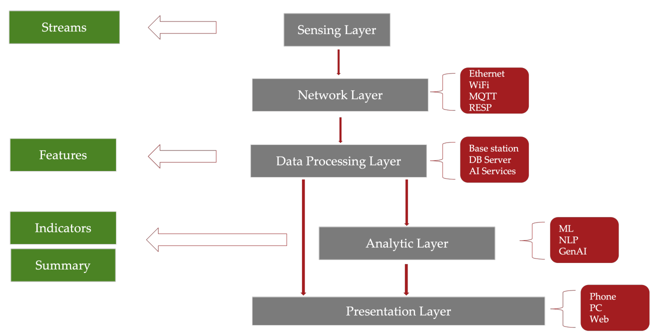


Figure 1: A multilayered IoT architecture. This structure comprises 1. Sensing Layer: responsible for data acquisition from various sensors. 2. Network Layer: facilitates data transmission using protocols such as Ethernet, WiFi, MQTT, and RESP. 3. Data Processing Layer: incorporates base stations, database servers, and AI services for initial data processing. 4. Analytic Layer: employs machine learning, natural language processing, generative AI, and other algorithms for advanced analytics 5. Presentation Layer: delivers results through various interfaces, including mobile phones, PCs, and web platforms.

Toolkit Design Principles: (1) **Modularity**: Components (*streams*, *bases*, *services*) are designed as loosely coupled, customizable modules; (2) **Extensibility**: Base classes are extendable to support new data sources and processing methods.

Platform Design Principles: (1) **Preserving Data Privacy**: The platform is engineered for local deployment without external internet communication during the runtime, reducing potential data exposure risks. Also, it allows users to choose between storing raw sensor data or only the derived feature data, thereby tailoring data retention policies for specific privacy requirements; (2) **Flexibility in Learning Context**: IoT concepts are incorporated

[1] https://github.com/ucph-ccs/OpenMMLA

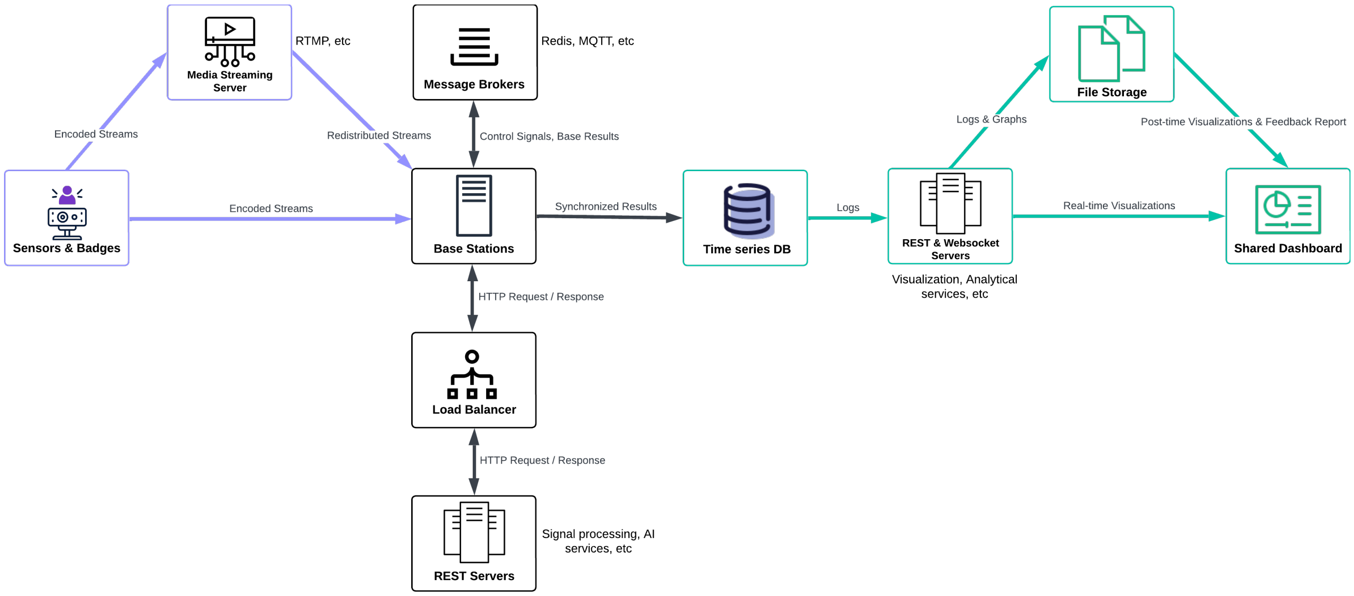


Figure 2: High-level system design with data flow. The platform's high-level design consists of three stages: input, processing, and output. In the data input stage, multimodal raw data from sensors & wearable badges are streamed directly to base stations or a central media server. The raw inputs are transformed into structured, coded streams for efficient transmission and processing. In the data processing stage, these encoded streams are processed individually by the corresponding *Base*, which handles some signal processing, while more complex tasks are offloaded to the server. For *Bases* within the same group, results are synchronized and uploaded to the time series database, where segment-level measurement features are generated. Finally, in the data output stage, these measurement features are visualized on the dashboard in real time and combined into indicators to analyze group interactions. The platform also generates post-processing visualizations, logs, and reports, which are stored and accessible via the shared dashboard, enabling both real-time awareness and retrospective analysis of group dynamics.

into the platform to accommodate diverse learning environments, such as laboratory settings that demand mobility. This approach enables data collection from geographically distributed teams at individual levels through wearable badges, enhancing the system's adaptability to the enormous scope of learning contexts; (3) **Scalability, Capacity, and Maintainability**: The platform employs a client-server architecture to enhance scalability, capacity, and maintainability. The bases/AIs and dashboard/analytics pairs (Figure 2) are aligned with this design approach, which divides computational tasks between clients and servers. This separation allows the platform to scale efficiently according to the number of participants needed for group analysis and handle concurrent requests from the dashboard clients. Still, it simplifies the management of the environment.

3.3 Implementation

The toolkit and platform are developed in Python and run on Unix-based systems (Mac/Linux/Raspberry Pi/WSL).

3.3.1 ***OpenMMLA Toolkit****.* The toolkit package provides programming interfaces fit for the platform components shown in Figure 2, encompassing *streams*, *bases*, *services*, and *utils* modules.

The *streams* module handles sensor data capture, transmission, and reception. Sensors transmit raw data to a base station or media server via wired or wireless communication. On the receiving side, *StreamReceiver* provides a unified interface to handle encoded streams from various sensors or media servers. It stores *StreamFrame* with acquisition timestamps into *StreamBuffer* in adjusting sampling rates to match the base station's processing speed. It also supports data fusion for streams with similar dimensions and ensures minimal data loss.

The *bases* module includes *Base* and *Synchronizer* classes. *Base* reads and processes frames from *StreamReceiver*, handling simple tasks like filtering and marker detection, while complex tasks are offloaded to REST servers. *Synchronizer* listens to MQTT channels, synchronizes and merges results from multiple sources, and generates group-level merged segment results.

The *services* module includes a set of computationally intensive algorithms for different streams (generating embeddings, noise reduction, image understanding, etc.) that can be deployed on environmentally independent REST servers, reducing operational conflicts with the base.

The *utils* module contains helper functions for the *bases* and *services* modules, including signal processing, server deployment, message brokers & database wrapping clients, logging, etc.

3.3.2 ***Platform Components: Sensors, Base Stations, Servers, Dashboard****.* The platform employs a diverse range of hardware and server architectures for various environments:

Sensors are divided into wearable badges and distributed environmental sensors. We designed three badge types: Voice Badge

(Nicla Vision board[2] with power supply), Regular Badge (AprilTag only [23]), and Vision Badge (Nicla Vision board with AprilTag and power supply) [19]. The Nicla Vision board integrates a camera, microphone, IMU, and Bluetooth/WiFi modules, enabling egocentric data collection. Environmental sensors include the Jabra Speak2 75 and Logitech C920 webcam (with Raspberry Pi) for large-area monitoring and contextual data capture.

Base stations process data streams using various hardware, from microprocessors to PCs. Each base station runs one or more instances of *Base* and *Synchronizer*, with specific types (e.g., *AudioSynchronizer*) synchronizing data from corresponding *Base* components (e.g., *AudioBase*).

Servers provides centralized services for other devices within distributed environments., including an InfluxDB time-series database, an RTMP streaming server, REST servers for AI services, a WebSocket server for real-time visualization, and an application server hosting the dashboard (React and Flask).

Dashboard are web-page interfaces accessible via phone and web browsers, featuring on session selection, real-time visualizations (e.g., speaker recognition, speech transcription), and post-time analysis (e.g., movement trajectory, interaction graphs). Measurement logs can be downloaded in JSON format.

3.3.3 ***Ready-to-use Pipelines.*** Leveraging the toolkit and grounded in learning theories [12], we developed a set of built-in data pipelines for the platform. These pipelines generate measurement features at different levels of granularity from multimodal data streams, including speaker ID, transcribed text, position, orientation, and action.

Real-time Audio Analyzer. In MMLA, audio is essential for understanding group communication. Figure 3 shows the implemented real-time audio analyzer, which reads continuous audio samples via *StreamReceiver* and processes them by segments. Initial processing includes decibel normalization, noise reduction (denoiser [10]), voice activity detection (silero[31]), and optional speech separation. Speaker embeddings are generated using the TitaNet-L model [17]. At enrollment, these embeddings create speaker profiles; during recognition, they are compared with enrolled speakers to identify those with the highest similarity and exceeding a threshold. If the current and previous speakers match, the segments are concatenated and transcribed by the Whisper model [26]. Segment length is adjustable, balancing recognition granularity and accuracy. The analyzer generates segment-level speaker recognition and chunk-level speech transcription, combined into ASR diarization logs.

Post-time Audio Analyzer. Figure 4 illustrates the post-time audio analyzer, which functions similarly to the real-time analyzer but processes entire audio files instead of streams. The analyzer splits the audio into fixed-length segments, performs speaker recognition chronologically, and labels each segment. Consecutive segments from the same speaker are aggregated into chunks and then transcribed. This provides an alternative method for generating speaker recognition and transcription features from traditional audio recordings.

Indoor Positioning. Alongside audio analysis, tracking participants' spatial movements and orientations in real time is essential for capturing the full scope of group dynamics. We implemented a multicamera indoor positioning pipeline to achieve this, enabling spatial tracking through cameras and AprilTag detection [23]. The setup requires AprilTag (a fiducial marker that can be used for a variety of tasks such as augmented reality, robotics, and camera calibration), cameras (e.g., Logitech C920 for capturing video frames), and multiprocessors (e.g., Raspberry Pi 5 for transmitting captured video frames). It generates spatial measurement features such as the position and orientation of the identified person under a unified coordinate system and the spatial relationships between people.

Video Frame Analyzer. For nonverbal behavior analysis, we utilized large language models (LLMs) and vision-language models (VLMs) for action recognition in 5. Traditionally, action recognition can be achieved through IMU sensor data and computer vision, with the latter generally offering higher accuracy. It often involves analyzing human interactions with surroundings, captured effectively by strategically placed cameras providing an observer's perspective. The video frame analyzer, implemented as a Flask application, processes image frames transmitted by *VideoBase* every 30 seconds. It first detects AprilTags and calculates their in-image coordinates, constructing the prompt for the VLM. The VLM links individuals with their ID and describes gestures, postures, and interactions. The predefined actions and image descriptions then assemble the prompt for LLM. Finally, the LLM classifies the described behaviors into predefined categories and parses them into a dictionary of *<ID, Action>* pairs.

3.3.4 ***Synchronization and Data Fusion.*** Synchronization ensures coherent and meaningful multimodal data stream analysis under a unified observing framework. *Spatial Synchronicity:* The indoor positioning system synchronizes multicamera streams by transforming each camera's coordinates into a shared reference coordinate system, enabling a cohesive spatial representation. *Temporal Synchronicity:* As for group analysis, individual time-segment results are aligned with group segments using strategies (earliest/nearest), making them temporally consistent for meaningful comparison and aggregation, thereby generating merged results that reflect the group dynamic.

Data fusion integrates information from various modalities to provide deeper insights. Before fusion, data must be temporally aligned to interpret cross-modal interactions accurately. *Feature-Level Fusion*: combines processed or low-level features extracted from raw signals. For instance, in the video frame analyzer, embeddings derived from image data and structured textual features (e.g., AprilTag coordinates or descriptive prompts) are fused to enhance context-aware action recognition. *Decision-Level Fusion*: operates on measurement features, which are derived outputs from specific data processing pipelines. For example, speaker recognition (audio) and body orientation/proximity (video) measurement features can be fused to infer directional communication patterns.

3.3.5 ***Realistic Considerations.*** The dynamics of real-world systems present a significant challenge to predictive machine learning models, as system changes over time can lead to performance degradation during the model's life cycle. OpenMMLA addresses

[2]https://docs.arduino.cc/hardware/nicla-vision/

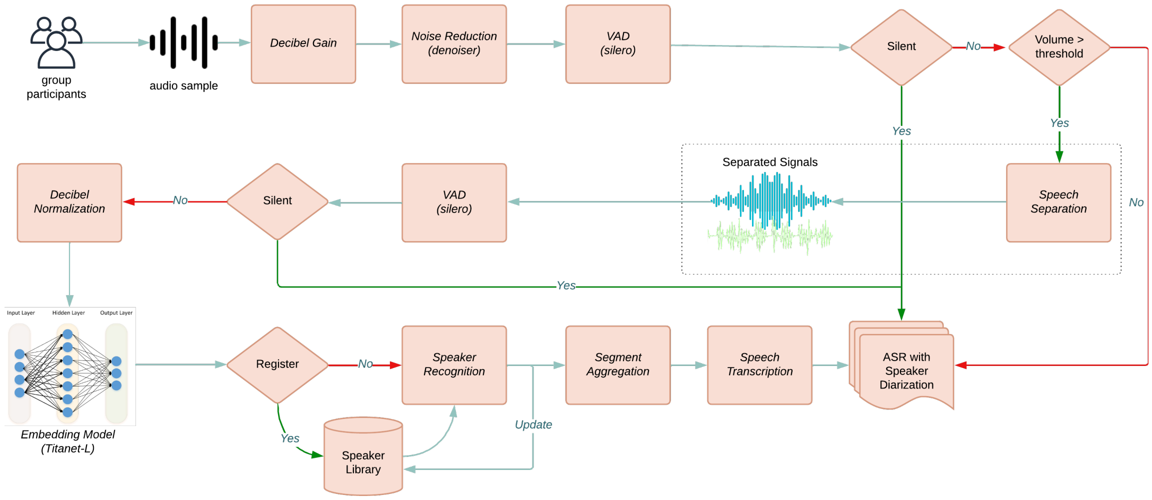


Figure 3: Real-time audio analyzer pipeline

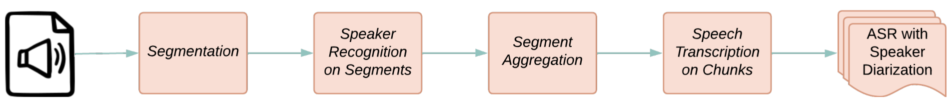


Figure 4: Post-time audio analyzer pipeline

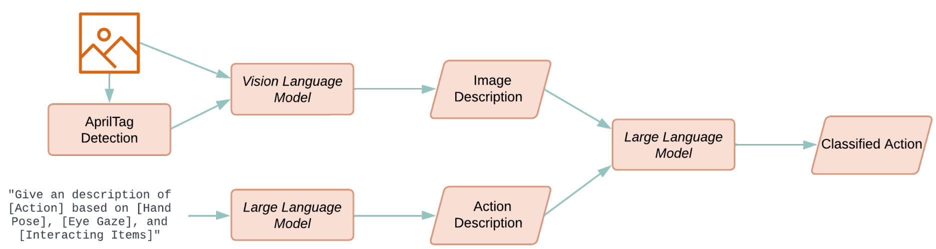


Figure 5: Video frame analyzer pipeline for action recognition

these challenges but requires further investigation in several key areas. Key considerations include the resolution of camera systems, where webcams are preferred over IP cameras due to higher resolution and privacy concerns. Additionally, depending on the task, data fusion strategies operate at the decision, feature, or, rarely, signal(stream) level. For instance, higher-order interactions like directional communication patterns are inferred by fusing audio-visual data at the decision level, while action recognition typically relies on feature-level fusion. Signal-level fusion—combining raw data such as waveforms or pixel grids—is theoretically possible but impractical due to the complexity of aligning diverse data types. An example might involve modifying pixel data by embedding identifiers (e.g., AprilTag metadata) directly onto the image to enhance comprehension by vision-language models.

4 System Evaluation

Data Capturing Test The synchronicity of different modalities is guaranteed by their sensor data acquisition time for the real world. Therefore, testing for clock drift during badge sampling is essential. The acquisition timestamp is based on the *time.ticks_us()* function with close precision to the CPU clock, and the periodically synced reference time from the NTP server reflected by the periodically falling blue lines in Figure 6 (left), which is calculated as ($Acquisition\ Time = Elapsed\ Time\ Ticks + Reference\ Time$). By comparing the data package's received timestamp with its acquisition timestamp, the difference between the two gives the network latency plus the amount of clock drift. The figure shows the time difference in blue and the variation in clock drift in green, calculated as the differences between consecutive time difference points while compensating for network latency. The green line remains nearly horizontal, indicating the clock drifting rate is nearly constant (~ 7 milliseconds/hour).

The data sampling rate of the badge's microphone sensor was tested via UDP transmission for 3600 seconds, as shown in Figure 6. The instantaneous average sampling rate, calculated by dividing the accrued number of data samples by the elapsed time, initially appears slower than the expected rate of 16000 Hz. This initial slowdown may be due to network latency or clock drift. While the

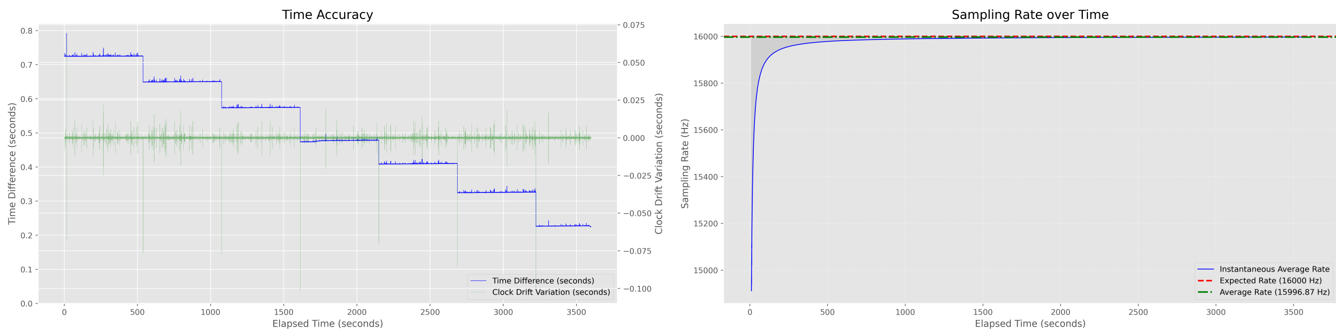
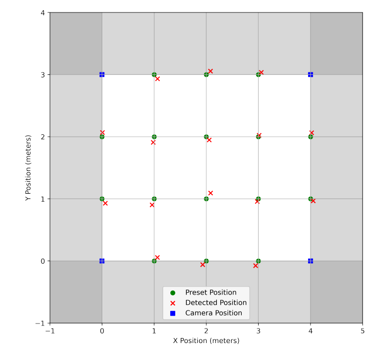


Figure 6: Time accuracy (left) and Sampling Rate over Time (right)



(a) 2D positioning plot (normal ambient light, no occlusion)

(b) Participants with regular badges interacting with the iPad. Predefined actions: {*Talking*: when a person is directly looking at others but not using any device, *Working-software*: when a person is using a tablet device like iPad and collaborating with others, *Working-hardware*: when a person is manipulating a microcontroller like Micro:bits and collaborating with others, *Distracted*: when a person is using a mobile phone, *Unclear*: you are not confident to categorize them into any of the above action classes}

Figure 7: Indoor positioning test (a) and Video frame analyzer test (b)

overall average sampling rate (15996.87 Hz) is close to the expected rate (16000 Hz), some discrepancies may be caused by data packet loss during transmission and fluctuations in the number of packets received in the last second. Specifically, 16000 samples (each 2 bytes) require 31.25 packets (each 1024 bytes) per second, but the actual number of packets received varies between 31 and 32.

The packet receiving rate was tested by transmitting audio packets from the badge over 4,253 seconds using UDP. Only 25 packets out of 132900 packets were lost, achieving a data receiving rate greater than 99.98%. This suggests that UDP is a reliable transport protocol for streaming packets for IoT-based data collection devices like badges. The bandwidth requirement for a combined audio and video stream from the badge is around 1 Mbps.

Data Processing Test To evaluate data processing pipelines, we conducted preliminary comparisons of the platform-generated results against human-annotated data as reference standards to see the deviations.

The audio analyzer pipeline was evaluated through two distinct trials: a 10-minute segment from a one-hour noisy meeting (n=4) recorded with a Jabra Speak2 75 device and a one-hour group seminar (n=6) wearing voice badges. The first trial achieved 18.5% diarization error rate, 27.9% word error rate, 27.3% match error rate, and 85.4% word coverage rate when compared to human-annotated data [20], and the second trial demonstrated strong alignment between sensor-measured and human-observed verbal interactions [15]. Additionally, we evaluated the indoor positioning pipeline's localization accuracy in a 4x3 meter space equipped with four Logitech C920 cameras. A human wearing a Tag Badge (61 mm) was detected by *VideoBases* at each preset location. Figure 7a illustrates the test results, showing an average distance error of ~8 cm. We tested the video frame analyzer using the VLM (MiniCPM-V-2_6) and LLM (Phi-3-small-128k-instruct) models, with parameters set to temperature = 0 and top_k = 1. Figure 7b presents the outputs, highlighting the pipeline's strong visual comprehension and inductive reasoning capabilities.

5 Conclusion, Discussion & Future Work

The paper introduces and provides an initial review of our Open-MMLA approach, beginning to answer the research aim of developing a multimodal data collection toolkit and platform to collect, analyze, and support small-group collaboration. The benefits of the toolkit approach introduced by [14] provide a forward direction for making a more industrial-strength platform that allows researchers to use different services that include broader generative AI [37] to capture, process, and analyze the relevant data from learning activities. Additionally, OpenMMLA provides tools that can be adapted to different environments and measures without being limited to a specific system or context [5] to improve generalizability. We also begin to address the design principles of data privacy, flexibility, and scalability by the development of the OpenMMLA platform. We achieve data privacy, for instance, by using badges while ensuring flexibility and scalability with our toolkit approach. By developing an open platform, additional sensors (physiological, wearable devices, etc) and environments can be used, providing new opportunities for research and collaboration. Besides, by providing a platform, the community can contribute to more general tools, allowing for more reproducibility and transparency [34], especially in providing tools that utilize AI for support [7].

Several areas require further exploration to enhance the current system. First, body orientation provides only indirect measures of attention and may need to be supplemented with other tracking methods like eye gaze. Additionally, we did not analyze the effect of the badges on user behavior. It is possible that individuals may not behave naturally while wearing badges, raising questions about whether the group work dynamics measured with badges would remain consistent without them. Also, the occlusion of the line of sight with the badges must be considered. Furthermore, while the badge is effective for streaming a single modality (either audio or video), it presents limited video quality due to the narrow-angle and compressed format used for transmission. Future work could explore replacing these badges with wearable devices like smartphones for more flexible and higher-quality data capture across multiple modalities. Developing more comprehensive analytical frameworks and conducting empirical studies would also enhance our understanding of collaborative learning processes and reveal pedagogical implications from OpenMMLA's outputs.

References

[1] Paulo Blikstein and Marcelo Worsley. 2016. Multimodal Learning Analytics and Education Data Mining: using computational technologies to measure complex learning tasks. *Journal of Learning Analytics* 3, 2 (2016), 220–238. https://doi.org/10.18608/jla.2016.32.11

[2] Dan Bohus, Sean Andrist, Ashley Feniello, Nick Saw, Mihai Jalobeanu, Patrick Sweeney, Anne Loomis Thompson, and Eric Horvitz. 2021. Platform for situated intelligence. *arXiv preprint arXiv:2103.15975* (2021).

[3] Oscar Canovas and Felix J. Garcia. 2023. *Analysis of Classroom Interaction Using Speaker Diarization and Discourse Features from Audio Recordings*. Springer International Publishing, 67–74. https://doi.org/10.1007/978-3-031-26190-9_7

[4] Pankaj Chejara, Reet Kasepalu, Luis Pablo Prieto, María Jesús Rodríguez-Triana, Adolfo Ruiz-Calleja, and Shashi Kant Shankar. 2023. Multimodal Learning Analytics Research in the Wild: Challenges and Their Potential Solutions.. In *CrossMMLA@ LAK*. 36–42.

[5] Pankaj Chejara, Luis P Prieto, Maria Jesus Rodriguez-Triana, Reet Kasepalu, Adolfo Ruiz-Calleja, and Shashi Kant Shankar. 2023. How to build more generalizable models for collaboration quality? lessons learned from exploring multi-context audio-log datasets using multimodal learning analytics. In *LAK23: 13th International Learning Analytics and Knowledge Conference*. 111–121.

[6] Pankaj Chejara, Luis P. Prieto, Adolfo Ruiz-Calleja, María Jesús Rodríguez-Triana, Shashi Kant Shankar, and Reet Kasepalu. 2020. *Quantifying Collaboration Quality in Face-to-Face Classroom Settings Using MMLA*. Springer International Publishing, 159–166. https://doi.org/10.1007/978-3-030-58157-2_11

[7] Mutlu Cukurova. 2024. The interplay of learning, analytics and artificial intelligence in education: A vision for hybrid intelligence. *British Journal of Educational Technology* (2024). https://doi.org/10.1111/bjet.13514

[8] Mutlu Cukurova, Michail Giannakos, and Roberto Martinez-Maldonado. 2020. The promise and challenges of multimodal learning analytics. *British Journal of Educational Technology* 51, 5 (2020), 1441–1449. https://doi.org/10.1111/bjet.13015

[9] Mutlu Cukurova, Rose Luckin, Eva Millán, and Manolis Mavrikis. 2018. The NISPI framework: Analysing collaborative problem-solving from students' physical interactions. *Computers & Education* 116 (2018), 93–109. https://doi.org/10.1016/j.compedu.2017.08.007

[10] Alexandre Defossez, Gabriel Synnaeve, and Yossi Adi. 2020. Real Time Speech Enhancement in the Waveform Domain. In *Interspeech*.

[11] Jon Echeverria and Olga C. Santos. 2021. KUMITRON: Artificial intelligence system to monitor karate fights that synchronize aerial images with physiological and inertial signals. In *26th International Conference on Intelligent User Interfaces-Companion*. 37–39.

[12] Michail Giannakos and Mutlu Cukurova. 2023. The role of learning theory in multimodal learning analytics. *British Journal of Educational Technology* 54, 5 (2023), 1246–1267. https://doi.org/10.1111/bjet.13320

[13] Arthur C. Graesser, Stephen M. Fiore, Samuel Greiff, Jessica Andrews-Todd, Peter W. Foltz, and Friedrich W. Hesse. 2018. Advancing the Science of Collaborative Problem Solving. *Psychological Science in the Public Interest* 19, 2 (2018), 59–92. https://doi.org/10.1177/1529100618808244%M30497346%Uhttps://journals.sagepub.com/doi/abs/10.1177/1529100618808244

[14] Javaria Hassan, Jovin Leong, and Bertrand Schneider. 2021. Multimodal Data Collection Made Easy: The EZ-MMLA Toolkit: A data collection website that provides educators and researchers with easy access to multimodal data streams.. In *LAK21: 11th international learning analytics and knowledge conference*. 579–585.

[15] Viktor Holm-Janas, Oriel Caro Miya Marshall, Zaibei Li, Jesper Bruun, and Daniel Spikol. 2024. Design Framework for Multimodal Learning Analytics Leveraging Human Observations. In *Technology Enhanced Learning for Inclusive and Equitable Quality Education*. Vol. 15160. Springer Nature Switzerland, Cham, 106–112. https://doi.org/10.1007/978-3-031-72312-4_13

[16] Sanna Järvelä, Dragan Gašević, Tapio Seppänen, Mykola Pechenizkiy, and Paul A. Kirschner. 2020. Bridging learning sciences, machine learning and affective computing for understanding cognition and affect in collaborative learning. *British Journal of Educational Technology* 51, 6 (2020), 2391–2406. https://doi.org/10.1111/bjet.12917

[17] Nithin Rao Koluguri, Taejin Park, and Boris Ginsburg. 2021. TitaNet: Neural Model for speaker representation with 1D Depth-wise separable convolutions and global context. arXiv:2110.04410 [eess.AS] https://arxiv.org/abs/2110.04410

[18] Oren Lederman, Akshay Mohan, Dan Calacci, and Alex Sandy Pentland. 2018. Rhythm: A Unified Measurement Platform for Human Organizations. *IEEE MultiMedia* 25, 1 (2018), 26–38. https://doi.org/10.1109/mmul.2018.112135958

[19] Zaibei Li, Martin Thoft Jensen, Alexander Nolte, and Daniel Spikol. 2024. Field report for Platform mBox: Designing an Open MMLA Platform. In *Proceedings of the 14th Learning Analytics and Knowledge Conference*. ACM, Kyoto Japan, 785–791. https://doi.org/10.1145/3636555.3636872

[20] Zaibei Li, Daniel Spikol, and Liv Nøhr. 2024. mBox-audio: Unveiling Conversational Dynamics through Real-Time and Post-Time Audio Analysis for MMLA. In *Companion Proceedings of the 14th International Conference on Learning Analytics and Knowledge (LAK24)*. 130–132. https://www.solaresearch.org/core/lak24-companion-proceedings/

[21] Xavier Ochoa. 2022. Multimodal systems for automated oral presentation feedback: A comparative analysis. In *The Multimodal Learning Analytics Handbook*. Springer, 53–78.

[22] Xavier Ochoa, Charles Lang, George Siemens, Alyssa Wise, Dragan Gasevic, and Agathe Merceron. 2022. Multimodal learning analytics-Rationale, process, examples, and direction. *The handbook of learning analytics* (2022), 54–65.

[23] Edwin Olson. 2011. AprilTag: A robust and flexible visual fiducial system. In *2011 IEEE International Conference on Robotics and Automation*. 3400–3407. https://doi.org/10.1109/ICRA.2011.5979561

[24] Sambit Praharaj, Maren Scheffel, Hendrik Drachsler, and Marcus Specht. 2021. Literature review on co-located collaboration modeling using multimodal learning analytics—Can we go the whole nine yards? *IEEE Transactions on Learning Technologies* 14, 3 (2021), 367–385.

[25] Samuel L. Pugh, Arjun Rao, Angela E.B. Stewart, and Sidney K. D'Mello. [n. d.]. Do Speech-Based Collaboration Analytics Generalize Across Task Contexts?. In *LAK22: 12th International Learning Analytics and Knowledge Conference*. ACM. https://doi.org/10.1145/3506860.3506894

[26] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2022. Robust Speech Recognition via Large-Scale Weak Supervision. arXiv:2212.04356 [eess.AS] https://arxiv.org/abs/2212.04356

[27] Luca Rossetto, Ivan Giangreco, Claudiu Tănase, and Heiko Schuldt. 2017. Multimodal video retrieval with the 2017 IMOTION system. In *Proceedings of the 2017 ACM on International Conference on Multimedia Retrieval*. 457–460.

[28] Nikol Rummel, Anne Deiglmayr, Hans Spada, George Kahrimanis, and Nikolaos Avouris. 2011. *Analyzing Collaborative Interactions Across Domains and Settings: An Adaptable Rating Scheme*. Springer US, 367–390. https://doi.org/10.1007/978-1-4419-7710-6_17

[29] Jan Schneider, Daniele Di Mitri, Bibeg Limbu, and Hendrik Drachsler. 2018. *Multimodal Learning Hub: A Tool for Capturing Customizable Multimodal Learning Experiences*. Springer International Publishing, 45–58. https://doi.org/10.1007/978-3-319-98572-5_4

[30] Caitlin Snyder, Nicole M Hutchins, Clayton Cohn, Joyce Horn Fonteles, and Gautam Biswas. 2024. Analyzing Students Collaborative Problem-Solving Behaviors

in Synergistic STEM+C Learning. In *Proceedings of the 14th Learning Analytics and Knowledge Conference*. ACM, 540–550. https://doi.org/10.1145/3636555.3636912

[31] Silero Team. 2024. Silero VAD: pre-trained enterprise-grade Voice Activity Detector (VAD), Number Detector and Language Classifier. https://github.com/snakers4/silero-vad.

[32] Hannah Vanderhoeven, Mariah Bradford, Changsoo Jung, Ibrahim Khebour, Kenneth Lai, James Pustejovsky, Nikhil Krishnaswamy, and Nathaniel Blanchard. 2024. *Multimodal Design for Interactive Collaborative Problem-Solving Support*. Springer Nature Switzerland, 60–80. https://doi.org/10.1007/978-3-031-60107-1_6

[33] Johannes Wagner, Florian Lingenfelser, Tobias Baur, Ionut Damian, Felix Kistler, and Elisabeth André. 2013. The social signal interpretation (SSI) framework: multimodal signal processing and recognition in real-time. In *Proceedings of the 21st ACM international conference on Multimedia*. 831–834.

[34] Marcelo Worsley, Roberto Martinez-Maldonado, and Cynthia D'Angelo. 2021. A New Era in Multimodal Learning Analytics: Twelve Core Commitments to Ground and Grow MMLA. *Journal of Learning Analytics* 8, 3 (2021), 10–27. https://doi.org/10.18608/jla.2021.7361

[35] Shunpei Yamaguchi, Motoki Nagano, Shunpei Ohira, Ritsuko Oshima, Jun Oshima, Takuya Fujihashi, Shunsuke Saruwatari, and Takashi Watanabe. 2022. Web Services for Collaboration Analysis With IoT Badges. *IEEE Access* 10 (2022), 121318–121328. https://doi.org/10.1109/access.2022.3222562

[36] Lixiang Yan, Vanessa Echeverria, Yueqiao Jin, Gloria Fernandez-Nieto, Linxuan Zhao, Xinyu Li, Riordan Alfredo, Zachari Swiecki, Dragan Gašević, and Roberto Martinez-Maldonado. 2024. Evidence-based multimodal learning analytics for feedback and reflection in collaborative learning. *British Journal of Educational Technology* 55, 5 (2024), 1900–1925.

[37] Lixiang Yan, Roberto Martinez-Maldonado, and Dragan Gasevic. 2024. Generative artificial intelligence in learning analytics: Contextualising opportunities and challenges through the learning analytics cycle. In *Proceedings of the 14th Learning Analytics and Knowledge Conference*. 101–111.

[38] Patrick H Zimmerman, J Elizabeth Bolhuis, Albert Willemsen, Erik S Meyer, and Lucas PJJ Noldus. 2009. The Observer XT: A tool for the integration and synchronization of multimodal signals. *Behavior research methods* 41 (2009), 731–735.